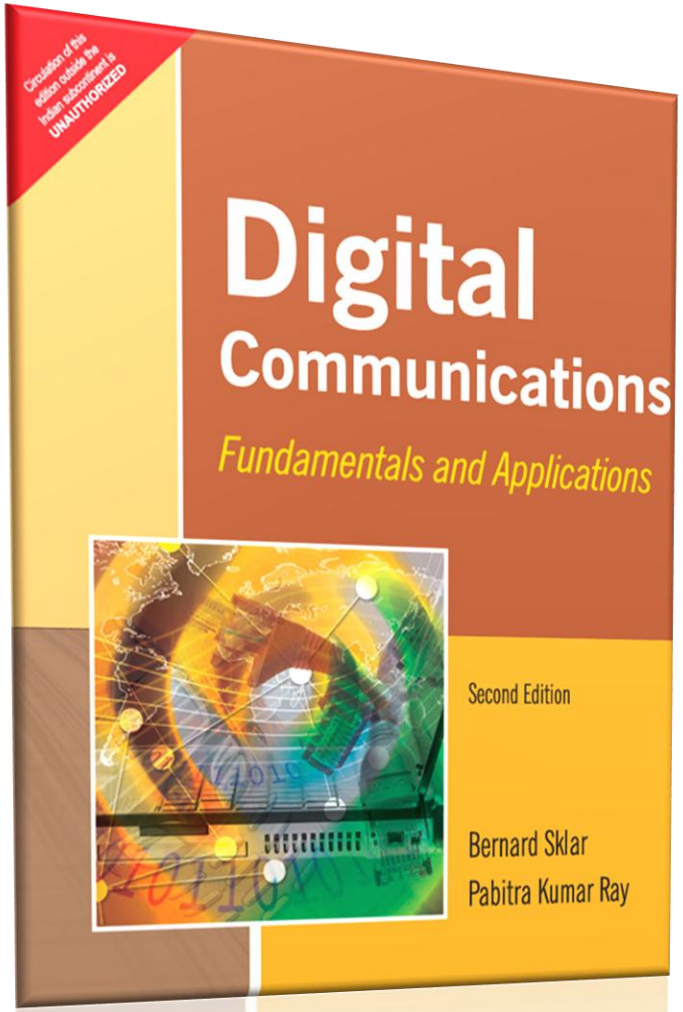


Digital communications: Fundamentals and Applications

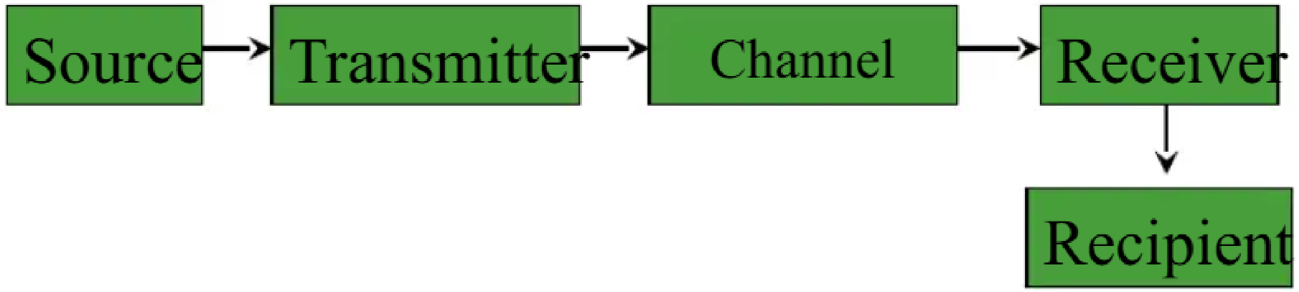
Presented By
U Yan Naung Soe CEIT (NSPU)

Question Bank :

Q-1 Explain basic communication system

Communication

- Main purpose of communication is to transfer information from a source to a recipient via a channel or medium.
- Basic block diagram of a communication system:



Brief Description

- **Source:** analog or digital
- **Transmitter:** transducer, amplifier, modulator, oscillator, power amp., antenna
- **Channel:** e.g. cable, optical fibre, free space
- **Receiver:** antenna, amplifier, demodulator, oscillator, power amplifier, transducer
- **Recipient:** e.g. person, (loud) speaker, computer

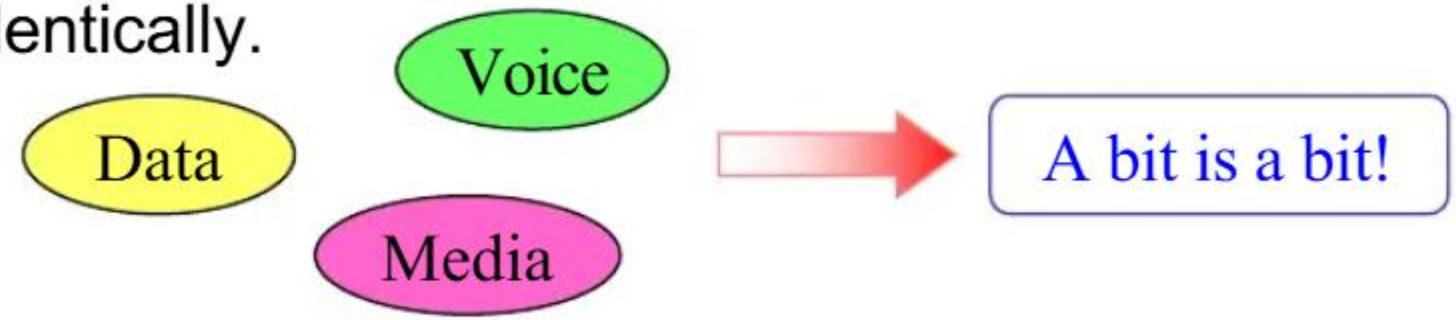
- **Types of information**
  Voice, data, video, music, email etc.

- **Types of communication systems**
  Public Switched Telephone Network (voice,fax,modem)
  Satellite systems
  Radio,TV broadcasting
  Cellular phones
  Computer networks (LANs, WANs, WLANs)

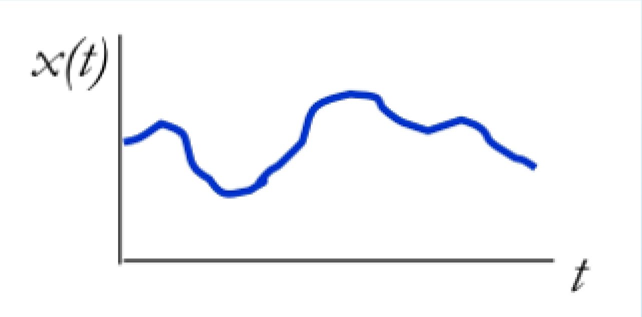
Digital versus analog

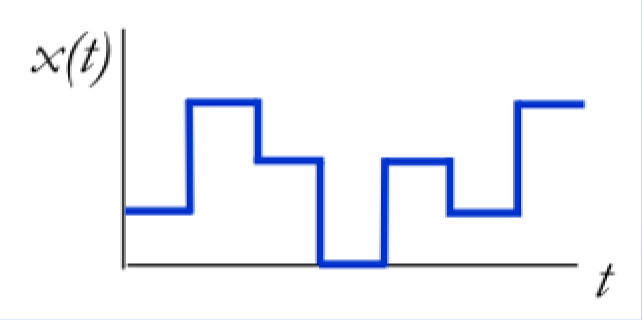
- Advantages of digital communications:
  - Regenerator receiver

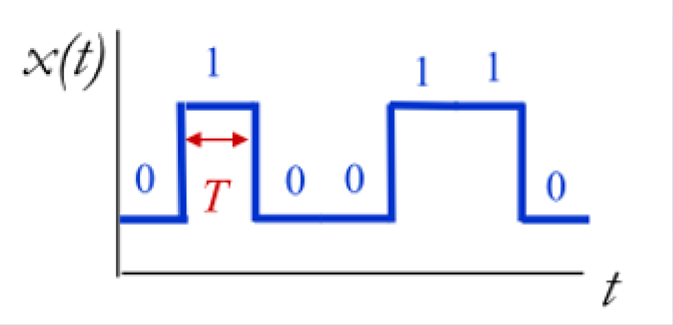
  - Different kinds of digital signal are treated identically.



- Analog signals
  - Value varies continuously
- Digital signals
  - Values limited to a finite set
- Binary signals
  - Two valued
  - Time T needed to send 1 bit
  - Data rate R=1/T bits per second







Q – 2 Why we use Digital Signal ?

Why digital?

- Digital techniques need to distinguish between discrete symbols allowing regeneration versus amplification

- Good processing techniques are available for digital signals, such as medium.
  - Data compression (or source coding)
  - Error Correction (or channel coding)(A/D conversion)
  - Equalization
  - Security

- Easy to mix signals and data using digital techniques

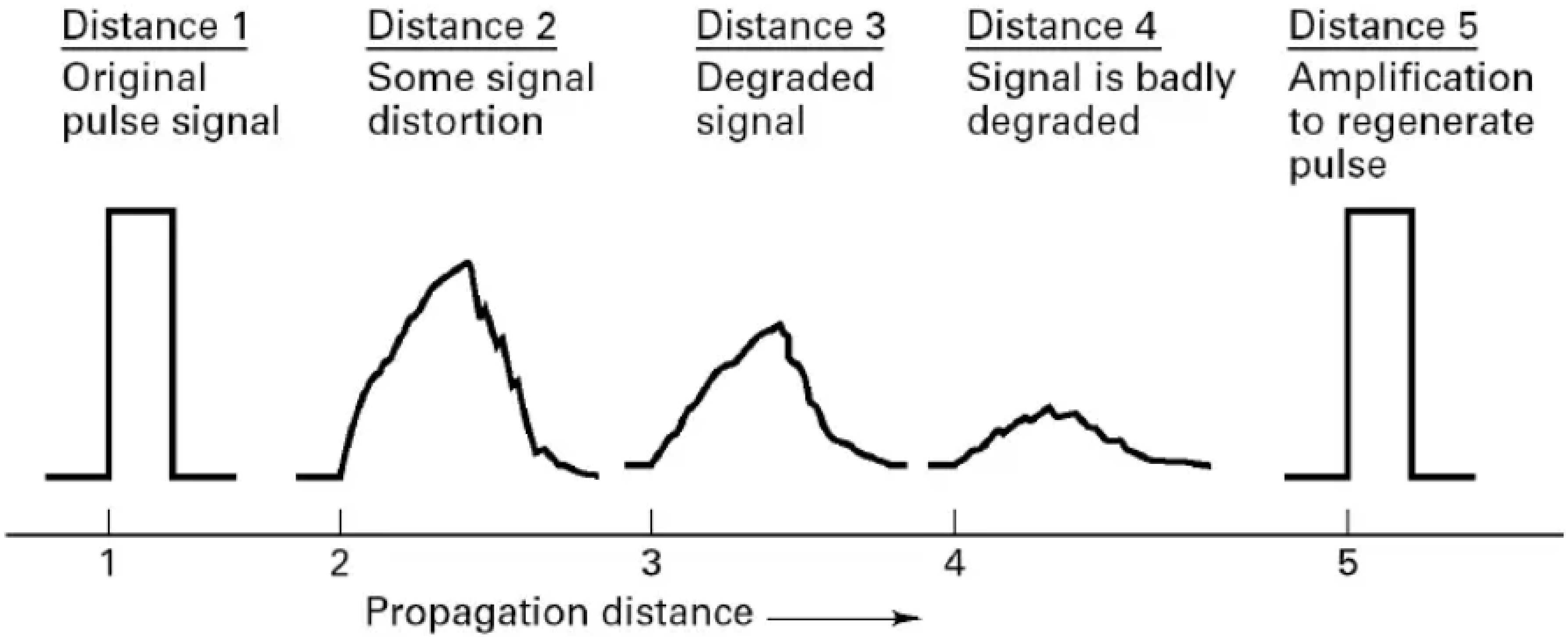


Figure 1.1 Pulse degradation and regeneration.

Q- 3 Draw the complete Digital communication system block diagram and explain each blocks in brief.

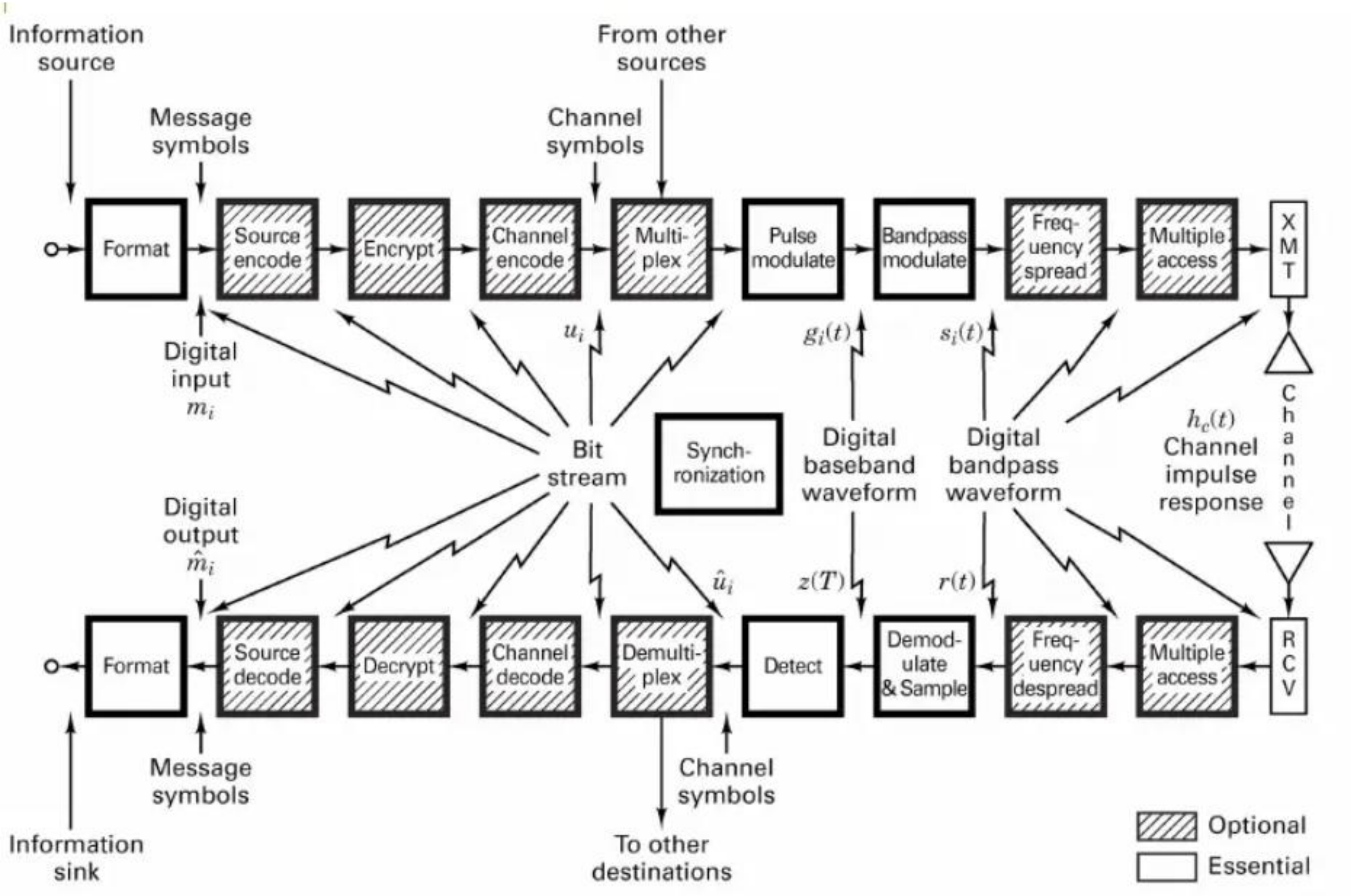


Figure 1.2 Block diagram of a typical digital communication system.

Q –5 Why we prefer Digital communication system ?

Why Digital Communications?

- Easy to regenerate the distorted signal
- Regenerative repeaters along the transmission path can detect a digital signal and retransmit a new, clean (noise free) signal
- These repeaters prevent accumulation of noise along the path

- This is not possible with analog communication systems
  - Two-state signal representation
- The input to a digital system is in the form of a sequence of bits (binary or M_ary)
  - Immunity to distortion and interference
  - Digital communication is rugged in the sense that it is more immune to channel noise and distortion

- Hardware is more flexible
- Digital hardware implementation is flexible and permits the use of microprocessors, mini-processors, digital switching and VLSI

- Shorter design and production cycle
  - Low cost
- The use of LSI and VLSI in the design of components and systems have resulted in lower cost
  - Easier and more efficient to multiplex several digital signals
  - Digital multiplexing techniques – Time & Code Division Multiple Access - are easier to implement than analog techniques such as Frequency Division Multiple Access

Q-6 Explain disadvantages of Digital communication system

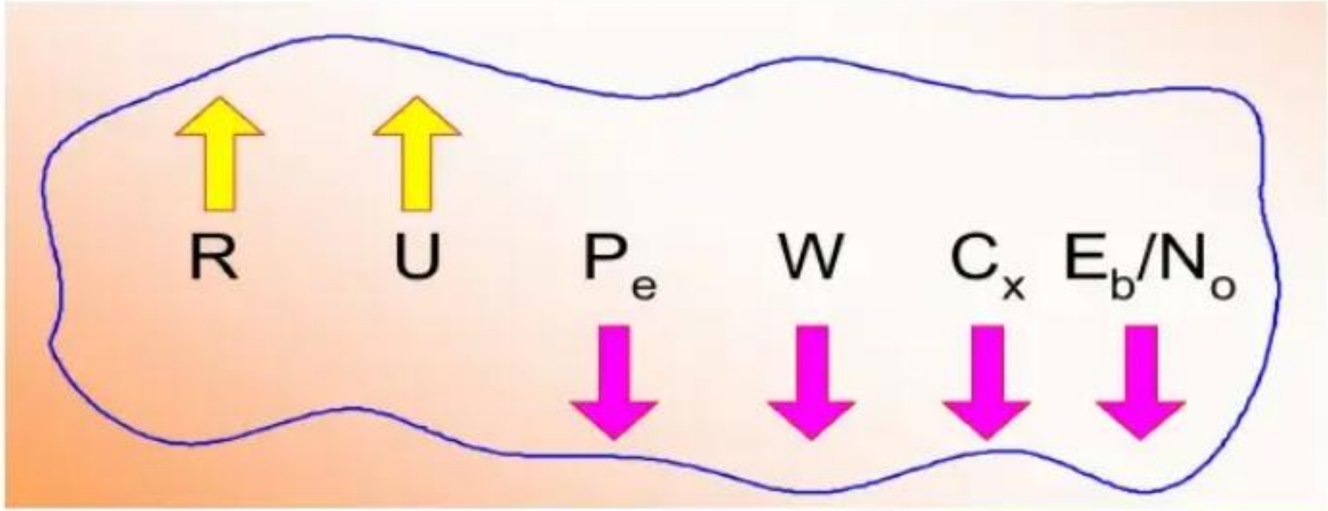
Disadvantages

- Requires reliable “synchronization”
- Requires A/D conversions at high rate
- Requires larger bandwidth
- Nongraceful degradation
- Performance Criteria
- Probability of error or Bit Error Rate

Q-7 Why designing digital communication system what care we must take ?

Goals in Communication System Design

- To maximize transmission rate, ***R***
- To maximize system utilization, ***U***
- To minimize bit error rate, P_e
- To minimize required systems bandwidth, ***W***
- To minimize system complexity, C_x
- To minimize required power, E_b/N_o



Q-8 Compare Digital and analog communication systems.

Q- 9 What is the advantage of digital over analog communication system ?

Comparative Analysis of Analog and Digital Communication

Digital Signal Nomenclature

- **Information Source**
  - Discrete output values e.g. Keyboard
  - Analog signal source e.g. output of a microphone
- **Character**
  - Member of an alphanumeric/symbol (A to Z, 0 to 9)
  - Characters can be mapped into a sequence of binary digits using one of the standardized codes such as
    - **ASCII: American Standard Code for Information Interchange**
    - **EBCDIC: Extended Binary Coded Decimal Interchange Code**

- **Digital Message**
  - Messages constructed from a finite number of symbols; e.g., printed language consists of 26 letters, 10 numbers, “space” and several punctuation marks. Hence a text is a digital message constructed from about 50 symbols
  - Morse-coded telegraph message is a digital message constructed from two symbols “**Mark**” and “**Space**”
- **M - ary**
  - A digital message constructed with *M* symbols
- **Digital Waveform**
  - Current or voltage waveform that represents a digital symbol
- **Bit Rate**
  - Actual rate at which information is transmitted per second

- **Baud Rate**
  - Refers to the rate at which the signaling elements are transmitted, i.e. number of signaling elements per second.

- **Bit Error Rate**
  - The probability that one of the bits is in error or simply the probability of error

Q-10 Classification of signals.

1.2 Classification Of Signals

1. Deterministic and Random Signals

- A signal is *deterministic* means that there is no uncertainty with respect to its value at any time.

- Deterministic waveforms are modeled by explicit mathematical expressions, example:

$$x(t) = 5\mathrm{Cos}(10t)$$

- A signal is *random* means that there is some degree of uncertainty before the signal actually occurs.

- Random waveforms/ Random processes when examined over a long period may exhibit certain regularities that can be described in terms of probabilities and statistical averages.

2. Periodic and Non-periodic Signals

- A signal x(t) is called *periodic in time* if there exists a constant T_0 > 0 such that

$$x(t) = x(t + T_0) \quad \text{for} \quad -\infty < t < \infty \qquad (1.2)$$

t denotes time

T_0 is the *period* of $x(t)$.

3. Analog and Discrete Signals

- An *analog signal* $x(t)$ is a continuous function of time; that is, $x(t)$ is uniquely defined for all t

- A *discrete signal* $x(kT)$ is one that exists only at discrete times; it is characterized by a sequence of numbers defined for each time, kT, where

  k is an integer

  T is a fixed time interval.

4. Energy and Power Signals

- The performance of a communication system depends on the received signal *energy;* higher energy signals are detected more reliably (with fewer errors) than are lower energy signals

- $x(t)$ is classified as an *energy signal* if, and only if, it has nonzero but finite energy ($0 < E_x < \infty$) for all time, where:

$$E_x = \lim_{T \to \infty} \int_{-T/2}^{T/2} x^2(t)\, dt \quad (1.7) \quad = \int_{-\infty}^{\infty} x^2(t)\, dt$$

- An energy signal has finite energy but *zero average power.*

- Signals that are both deterministic and non-periodic are classified as energy signals

4. Energy and Power Signals

- *Power* is the *rate* at which energy is delivered.

- A signal is defined as a power signal if, and only if, it has finite but nonzero power ($0 < P_x < \infty$) for all time, where

$$P_x = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\, dt \tag{1.8}$$

- Power signal has finite average power but *infinite energy.*

- As a general rule, periodic signals and random signals are classified as power signals

5. The Unit Impulse Function

- *Dirac delta function* $\delta(t)$ or impulse function is an abstraction—an infinitely large amplitude pulse, with zero pulse width, and unity weight (area under the pulse), concentrated at the point where its argument is zero.

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1 \qquad (1.9)$$

$$\delta(t) = 0 \quad \text{for} \quad t \neq 0 \qquad (1.10)$$

$$\delta(t) \text{ is bounded at } t = 0 \qquad (1.11)$$

- Sifting or Sampling Property

$$\int_{-\infty}^{\infty} x(t)\, \delta(t - t_0)\, dt = x(t_0) \qquad (1.12)$$

Q -11 Define Spectral Density

1.3 Spectral Density

- The *spectral density* of a signal characterizes the distribution of the signal's energy or power in the frequency domain.

- This concept is particularly important when considering filtering in communication systems while evaluating the signal and noise at the filter output.

- The energy spectral density (ESD) or the power spectral density (PSD) is used in the evaluation.

1. Energy Spectral Density (ESD)

- Energy spectral density describes the signal energy per unit bandwidth measured in joules/hertz.
- Represented as $\psi_x(f)$, the squared magnitude spectrum

$$\psi_x(f) = |X(f)|^2 \quad (1.14)$$

- According to Parseval's theorem, the energy of x(t):

$$E_x = \int_{-\infty}^{\infty} x^2(t)\,dt = \int_{-\infty}^{\infty} |X(f)|^2\,df \quad (1.13)$$

- Therefore:

$$E_x = \int_{-\infty}^{\infty} \psi_x(f)\,df \quad (1.15)$$

- The Energy spectral density is symmetrical in frequency about origin and total energy of the signal x(t) can be expressed as:

$$E_x = 2\int_{0}^{\infty} \psi_x(f)\,df \quad (1.16)$$

2. Power Spectral Density (PSD)

- The *power spectral density* (PSD) function $G_x(f)$ of the periodic signal $x(t)$ is a real, even, and nonnegative function of frequency that gives the distribution of the power of $x(t)$ in the frequency domain.
- PSD is represented as:

$$G_x(f) = \sum_{n=-\infty}^{\infty} |C_n|^2 \delta(f - nf_0) \qquad (1.18)$$

- Whereas the average power of a periodic signal x(t) is represented as:

$$P_x = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x^2(t)\,dt = \sum_{n=-\infty}^{\infty} |C_n|^2 \qquad (1.17)$$

- Using PSD, the average normalized power of a real-valued signal is represented as:

$$P_x = \int_{-\infty}^{\infty} G_x(f)\,df = 2\int_{0}^{\infty} G_x(f)\,df \qquad (1.19)$$

Q-12 Explain Autocorrelation

1.4 Autocorrelation

1. Autocorrelation of an Energy Signal

- Correlation is a matching process; *autocorrelation* refers to the matching of a signal with a delayed version of itself.
- Autocorrelation function of a real-valued energy signal $x(t)$ is defined as:

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t)\, x(t+\tau)\, dt \quad (1.21) \quad \text{for} \quad -\infty < \tau < \infty$$

- The autocorrelation function $R_x(\tau)$ provides a measure of how closely the signal matches a copy of itself as the copy is shifted τ units in time.
- $R_x(\tau)$ is not a function of time; it is only a function of the time difference τ between the waveform and its shifted copy.

1. Autocorrelation of an Energy Signal

- The autocorrelation function of a real-valued *energy* signal has the following properties:

$$R_x(\tau) = R_x(-\tau)$$ symmetrical in about zero

$$R_x(\tau) \le R_x(0) \text{ for all } \tau$$ maximum value occurs at the origin

$$R_x(\tau) \leftrightarrow \psi_x(f)$$ autocorrelation and ESD form a Fourier transform pair, as designated by the double-headed arrows

value at the origin is equal to the energy of the signal

$$R_x(0) = \int_{-\infty}^{\infty} x^2(t)\,dt$$

2. Autocorrelation of a Power Signal

- Autocorrelation function of a real-valued power signal *x*(*t*) is defined as:

$$R_x(\tau) = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, x(t+\tau)\, dt \quad \text{for} \quad -\infty < \tau < \infty \qquad (1.22)$$

- When the power signal *x*(*t*) is periodic with period T_0, the autocorrelation function can be expressed as

$$R_x(\tau) = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)\, x(t+\tau)\, dt \quad \text{for} \quad -\infty < \tau < \infty \qquad (1.23)$$